Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST ANDREW'S JUNIOR COLLEGE
2013 JC2 Preliminary Examinations

H2 BIOLOGY **9648/3**

Paper 3: Applications and Planning Question

Wednesday 18 September 2013 2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your civics group, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,
1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Paper 3	
1	/14
2	/14
3	/12
5	/20
Total	/60
4 (Planning)	/12

This document consists of **15** printed pages.

[Turn over

Answer **all** *questions*.

1. A student researcher attempted to clone the Human Growth Hormone (HGH) gene. Two separate samples of HGH DNA were obtained from DNA libraries and subjected to agarose gel electrophoresis. Results were shown in Fig.1.1.

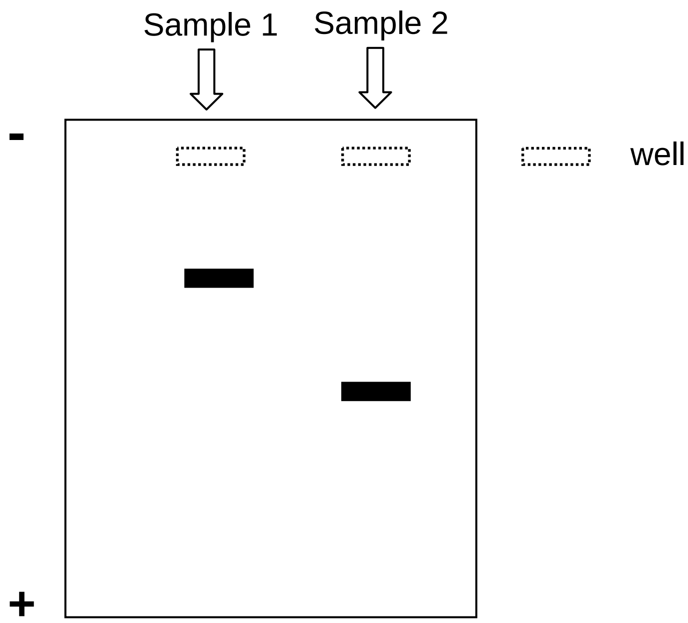

Fig. 1.1

(a)(i) Describe how the DNA fragments were separated into visible bands.

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Subsequent cloning procedures involving *Escherichia coli* host cells yield functional HGH protein when Sample 2 was used, but not for Sample 1.

(ii) Based on the difference in the bands observed in Fig. 1.1, account for the failure to produce functional HGH protein for Sample 1.

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(iii) State the type of DNA library where Sample 2 was obtained from.

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The student researcher has designed the following plasmid (Fig. 1.2) for use in her preliminary trials of a cloning experiment.

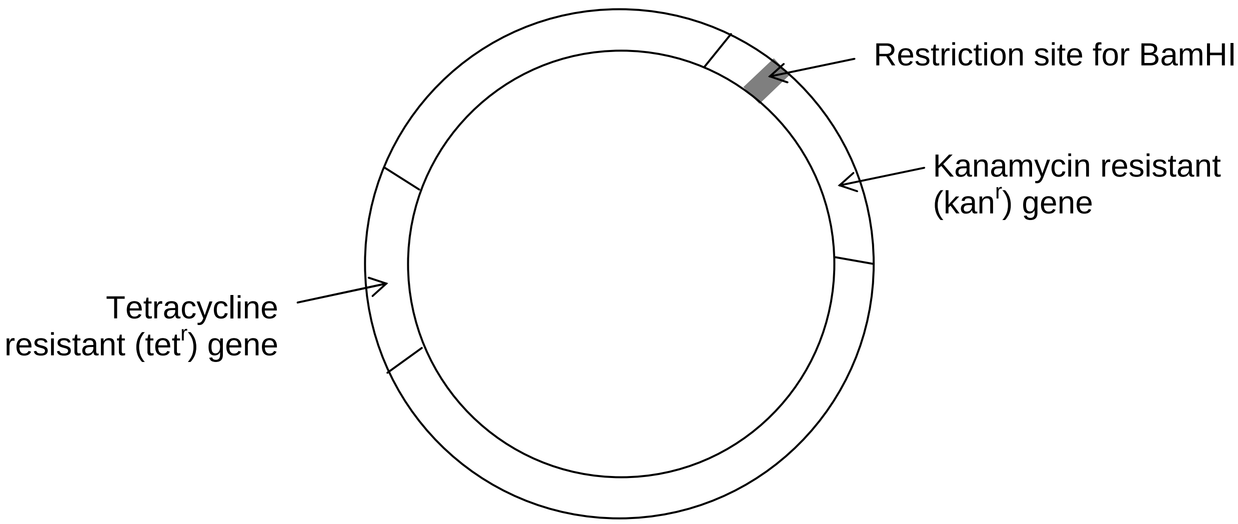

Fig. 1.2

(b) With reference to the properties of bacterial plasmids, provide 2 reasons why this plasmid in Fig 1.2 will not work well as a cloning vector.

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(c) The student researcher has since made improvements to the plasmid in Fig. 1.2 such that it is now a functional cloning vector.

Steps were then taken to create a recombinant plasmid using BamHI and the improved plasmid.

Restriction enzyme	Specific recognition site
BamHI	5'-G^G A T C C-3' 3'-C C T A G^G-5'

^ indicates where the restriction enzyme cuts

(i) Describe the natural function of BamHI restriction enzyme.

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(ii) Explain the main problem with using Sample 2 and BamHI in the formation of a recombinant plasmid and suggest how this problem can be solved.

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(iii) Describe the subsequent steps in the formation of recombinant plasmid.

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(iv) With reference to Fig. 1.2, explain how you would identify transformed bacterial cells that have taken up the recombinant plasmid with the HGH gene using antibiotic selection.

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[Q1 Total: 14]

2. Severe combined immunodeficiency (SCID) is a genetic disorder characterized by lack of functional lymphocytes, resulting in defects in both T and B cells responses of the immune system**.** Patients are susceptible to opportunistic infections while children usually die if not treated.

Fig. 2.1 shows the inheritance of SCID in a family in which both parents are carriers. Only one of the sons suffer from SCID.

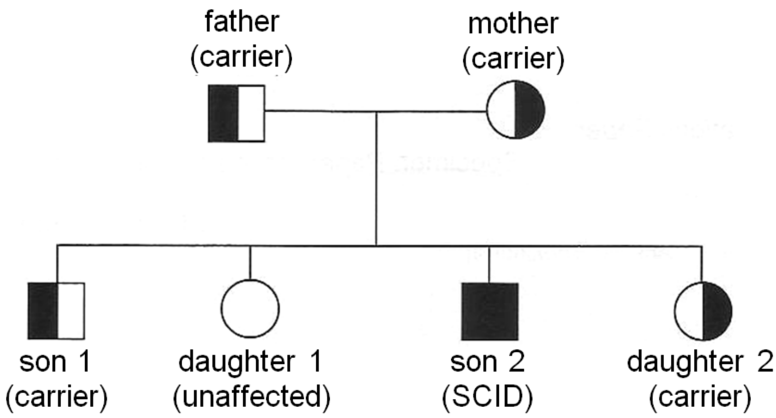

Fig. 2.1

(a) State the form of SCID that is being inherited in the family and the mode of inheritance.

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(b) Two RFLP morphs have been found tightly linked to the adenosine deaminase (ADA) gene which can be used for genetic screening. A radioactive probe has been designed to reveal the two RFLP morphs.

Both RFLP morphs and radioactive probe are shown in Fig. 2.1 with the arrows representing the *Pvu*II restriction sites. The asterisk (*) indicates the position of a mutation at the restriction site that results in the loss of the restriction site. Morph 2 was found to be linked to the mutant ADA gene.

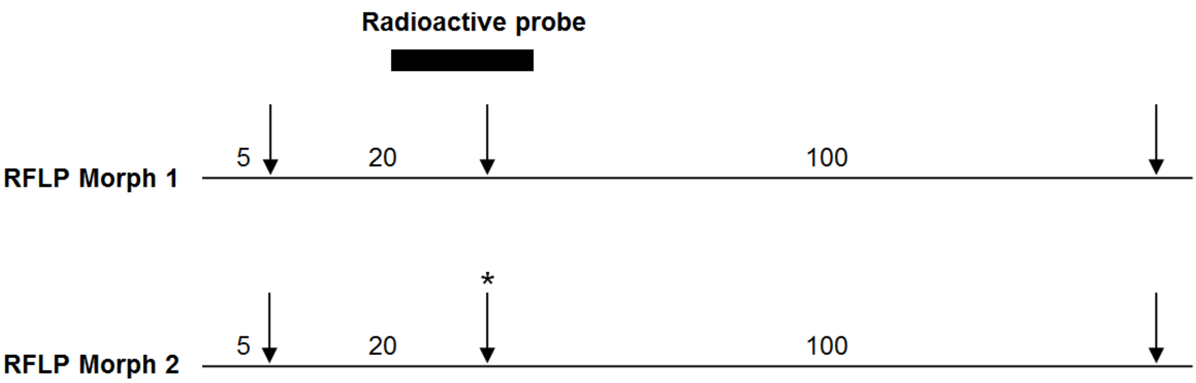
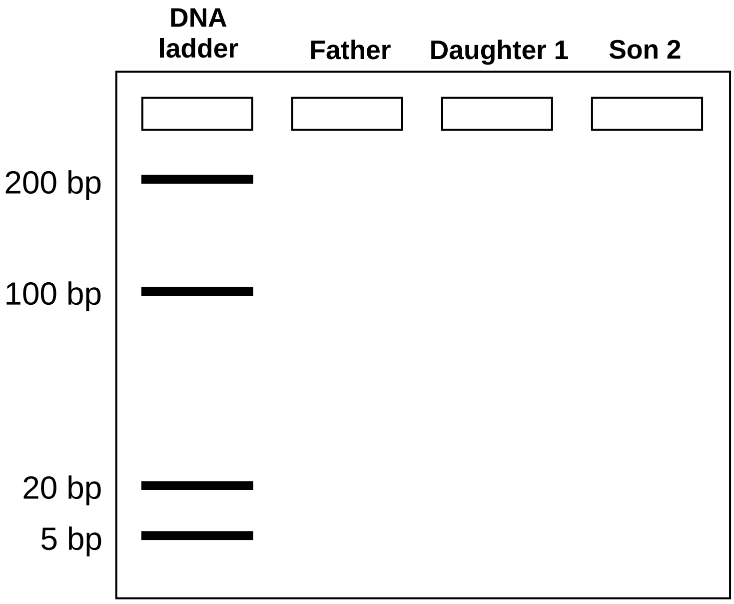


Fig. 2.2

(i) With reference to Fig 2.1 and Fig. 2.2, draw the resulting autoradiogram you would expect to see in the space below. [3]



(ii) Explain the role of the DNA ladder.

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(c) Apart from disease detection, other applications of RFLP include genetic fingerprinting and in genomic mapping.

(i) Explain why genes especially protein-encoding genes are seldom used as markers for genetic fingerprinting.

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(ii) Explain the use of genetic markers in genomic mapping in terms of linkage mapping.

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(d) Another method to detect ADA-SCID is through the use of PCR. Fig 2.3 shows the number of DNA molecules made using PCR, starting with one molecule.

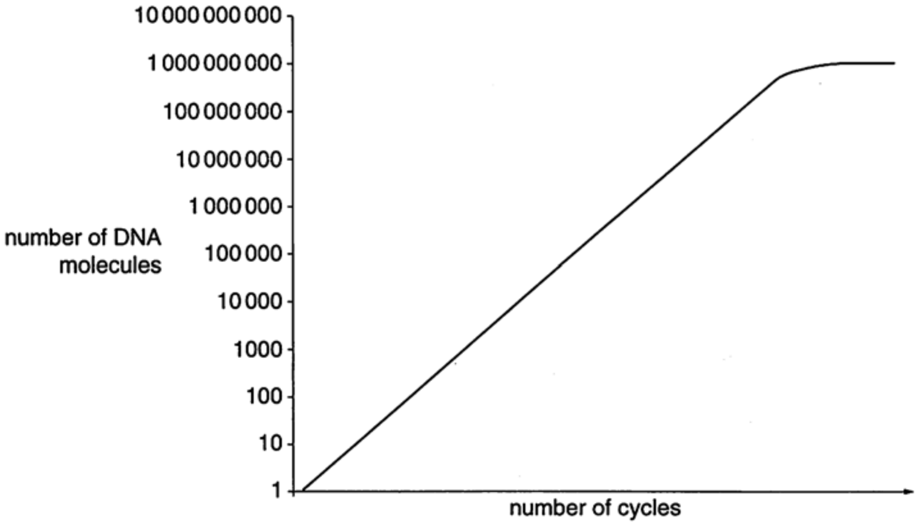

Fig 2.3

(i) Explain three advantages of PCR.

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(ii) Suggest two reasons for the leveling off of the graph from cycles 17 to 20.

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[Q2 Total: 14]

3.

(a) Many varieties of rice, *Oryza sativa*, die quickly when totally submerged by flood water during the monsoon season. Typically, submergence-intolerant plants respond to submerged conditions by elongating rapidly.

In 2006, a gene, *Sub1A*, on rice chromosome 9 was found to be involved in tolerance of prolonged submergence.

An allele, *Sub1A-1*, was found only in submergence-tolerant rice. Varieties that are submergence-intolerant either have a second allele, *Sub1A-2*, or lack the gene altogether.

(i) *Sub1A-1* codes for a protein that controls the transcription of the gene for alcohol dehydrogenase. This gene is activated when the plant is in anaerobic conditions.

Explain the importance to submerged rice plants of producing alcohol dehydrogenase.

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A variety of submergence-intolerant rice, which lacked the gene *Sub1A*, was genetically engineered to express *Sub1A-1.*

The mean heights of three types of rice plants were compared before and after 10 days of submergence in water. The types of rice plants were:

I : submergence-intolerant plants
I$^+$: submergence-intolerant plants genetically engineered to express *Sub1A-1*
T : submergence-tolerant plants

The results are shown in Fig. 3.1.

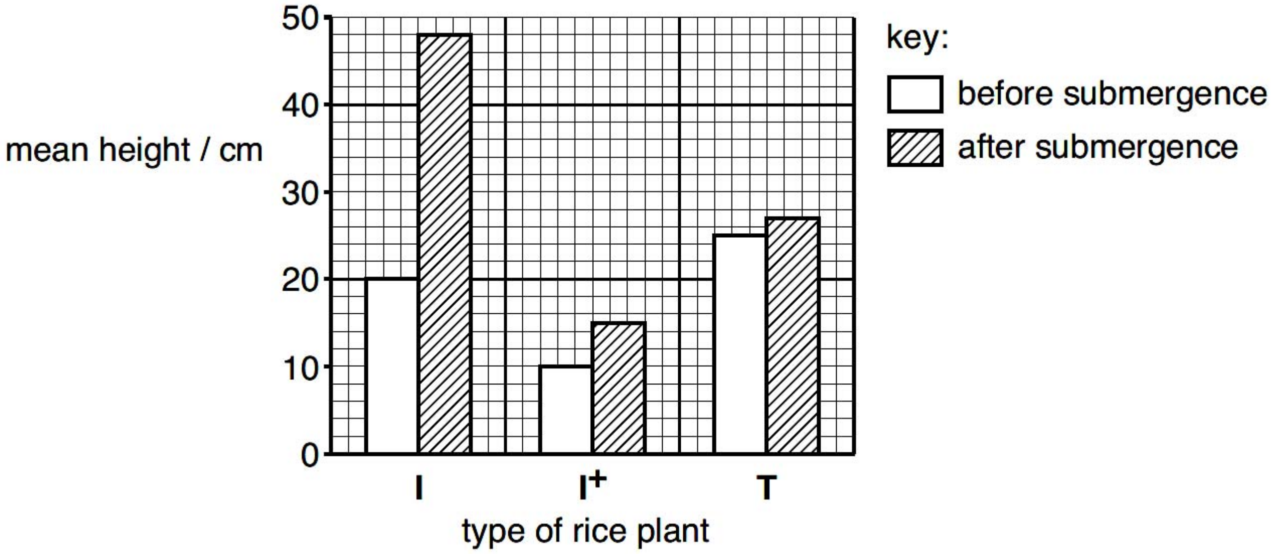

Fig. 3.1

(ii) With reference to Figure 3.1, compare the effect of submergence on the elongation of the three types of rice plants.

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The allele *Sub1A-1* could be introduced into high-yielding variety of submergence-intolerant rice either by genetic engineering or by selective breeding.

(iii) Suggest two advantages of using genetic engineering, rather than selective breeding, to introduce the allele.

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(b) Round-Up Ready^{TM} soybeans are genetically modified to be resistant towards Round-Up^{TM} herbicide, a glyphosate-based herbicide that normally inhibits an important enzyme in the aromatic amino acid synthesis pathway in plants. This allows farmers to freely apply Round-Up^{TM} to destroy weeds without affecting the Round-Up Ready^{TM} soybeans.

Round-Up Ready^{TM} soybeans may be produced by first transfecting soybean callus cells with the resistance gene from *Agrobacterium*.

(i) To obtain a callus culture, suitable explants must first be chosen. Suggest a suitable explant and explain your choice.

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(ii) Explain three advantages of growing soyabean plants from tissue culture rather than from seeds.

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(iii) Soyabean plants can reproduce sexually by means of seeds. Soyabean plants grown from seeds are very variable in their yield. Explain why.

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Genetically modifying soybean plants to be resistant towards Round-Up™ herbicide might threaten the environment and human safety.

(iv) State one possible adverse effect on:

The environment

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Human safety

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[Q3 Total: 12]

4. Planning question

The goshawk is an endangered species protected by the Convention of International Trade in Endangered Species of Wild Fauna and Flora (CITES).

The Royal Society for the Protection of Birds (RSPB) has found a man in possession of four young goshawks and an adult female. He claimed to have bred the young birds using his adult female and a male bird borrowed from another keeper of these birds of prey. The RSPB suspects that the young goshawks were poached from the wild and not bred in captivity as claimed by the man.

Plan an investigation using DNA fingerprinting to verify if the young goshawks were bred in captivity or poached from the wild.

[Total: 12]

You planning must be based on the assumption that you have been provided with the following equipment and materials

- tissue samples (feather tips) from the four young gohawks, the adult female and the adult male under investigation
- pestle and mortar
- DNA extraction buffer solution
- ice-cold ethanol
- glass rods
- microcentrifuge tubes
- centrifuge
- restriction enzyme
- PCR reagents and equipment
- agarose gel electrophoresis reagents and equipment
- suitable source of electrical current
- radioactive probe
- nitrocellulose membrane
- autoradiography equipment

Your plan should have a clear and helpful structure to include

- an explanation of the theory to support your practical procedure

- a description of the method used, including the scientific reasoning behind the method
- the type of data generated by the experiment
- how the results will be analysed including how the origin of the organism can be determined.
- relevant risks and precautions taken

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Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Compare the properties, features and functions of zygotic and embryonic stem cells. [5]

(b) Using one named example of an adult stem cell, describe its normal role and how it could be used in gene therapy against SCID using a retrovirus. [8]

(c) Discuss the social and ethical considerations for the use of gene therapy. [7]

[Total: 20]